

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

UNIT I (ELECTRO-PHYSIOLOGY AND BIO-POTENTIAL RECORDING)

Degree, Semester & Branch: V Semester B.E. ECE B

Course Code & Title: EC8073 Medical Electronics

Name of the Faculty member: Mr.S.Vijayakumar

Name of the Topic: EMG- lead systems

Name of the Innovative Practice: One minute paper

Date & Duration: 12.07.2019& 1 minute

ICT Tool Used:- -

Description:

A “one-minute paper” may be defined as a very short, in-class writing activity (taking one-minute or less to complete) in response to an instructor-posed question, which prompts students to reflect on the day’s lesson and provides the instructor with useful feedback.

Goals (Learning Outcomes):

- Students will be able to reflect on their understandings and attitudes after completing the Learning experiences from a unit or focus area.
- Students will be able to explain the operation of EMG- lead systems.

Use of appropriate methods:

Justification for choosing the topic using One minute paper activity:

Electromyography (EMG) measures muscle response or electrical activity in response to a nerve’s stimulation of the muscle. The test is used to help detect neuromuscular abnormalities. During the test, one or more small needles (also called electrodes) are inserted through the skin into the muscle. The electrical activity picked up by the electrodes is then displayed on an oscilloscope (a monitor that displays electrical activity in the form of waves). I choose this activity, whether my concept and what the students perceived as the main concept are the same. So, I asked to students to write EMG after teaching in a piece of paper.

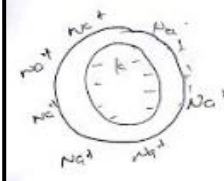
Effective presentation

Implementation (Plan & Execution) with Proof:

One minute papers is used at the end of a class to have students reflect back and think more deeply about the most important concept discussed in class that day. I asked the students must first organize their thinking to rank the major points and then decide upon a significant question. Students write a one-minute paper in response to my question. After one minute duration, I collected the papers from students.

Innovative Teaching Method Execution

EMG- lead systems :- One minute paper

M. Panshula 95361710605 Polarization: More positive charges are outside and negative charge are inside. Action Potential → Transfer of positive charge from outside to inside and π. Sodium pump. (-60 to -100) Depolarization. → Repolarization. → Back to its resting position. ↓ +20 -60 +40.	Medical Electronics. 953617106083 Resting and action potential 1. Polarization 2. Depolarization 3. Repolarization 4. all or nothing law 5. Sodium pump 7. action 
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Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- After conducting this activity I found that many students understand this topic.
- The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correctly.

Reflective Critique:

• **Benefits:**

Minute papers are an effective way of involving all students in class simultaneously. It ensures equal participation of each and every class member, including anyone who may be too shy or fearful to participate orally. It can promote class attendance and attentiveness. It can improve the quality of class discussion by having students write briefly about a concept or issue before they begin discussing it.

• **Challenges:**

After teaching this particular topic students asked some time to prepare. So, I gave few minutes to prepare which takes more time to implement this activity than planned.

Changes required for future (if required):Nil

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	2	-	-	-	-	1	--	-	-	1

CO1: The students will be able to explain about the physiological parameters and recording methods

References:

1. <https://oncourseworkshop.com/self-awareness/one-minute-paper/>
2. <https://www.humber.ca/centreforteachingandlearning/instructional-strategies/teaching-methods/classroom-management-designing-the-environment/feedback-in-the-classroom/one-minute-papers.html>
3. M. Arumugam, "Biomedical Instrumentation", Second Edition.

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Innovative Practices Description

UNIT II (BIO-CHEMICAL AND NON-ELECTRICAL PARAMETER MEASUREMENTT)

Degree, Semester & Branch: V Semester B.E. ECE B

Course Code & Title: EC8073 Medical Electronics

Name of the Faculty member: Mr.S.Vijayakumar

Name of the Topic: Blood Cell Counters

Name of the Innovative Practice: One minute paper

Date & Duration: 02.08.2019 & 1 minute

ICT Tool Used:- -

Description:

A “one-minute paper” may be defined as a very short, in-class writing activity (taking one-minute or less to complete) in response to an instructor-posed question, which prompts students to reflect on the day’s lesson and provides the instructor with useful feedback.

Goals (Learning Outcomes):

- Students will be able to reflect on their understandings and attitudes after completing the Learning experiences from a unit or focus area.
- Students will be able to explain the operation of blood cell counter.

Use of appropriate methods:

Justification for choosing the topic using One minute paper activity:

Blood cell counter is used to measure the blood cells counts in blood sample. There are different types of methods to measure the blood cell counts. I choose this activity, whether my concept and what the students perceived as the main concept are the same. So, I asked to students to write blood cell counter after teaching in a piece of paper.

Effective presentation

Implementation(Plan & Execution) with Proof:

One minute papers is used at the end of a class to have students reflect back and think more deeply about the most important concept discussed in class that day. I asked the students must first organize their thinking to rank the major points and then decide upon a significant question.

Students write a one-minute paper in response to my question. After one minute duration, I collected the papers from students.

Innovative Teaching Method Execution	
Blood Cell Counters :- One minute paper	
953617106089 III - year ECE-B. Blood cell counters - counting the red blood cells and white blood cells, PCV - packed cell volume. Method - microscopic method automatic optical method electrical conductivity method	Reg no: 953617106089 III year ECE-B 1) Coulter counter 2) Erythrocytes 3) Thrombocytes - blood coagulation 4) Leucocytes - immunity 5) Microscopic method 6) Automatic 7) Electrical conductivity method 8) Temperature measurements

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- After conducting this activity I found that many students understand this topic.
- The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correctly.

Reflective Critique:

- **Benefits:**

Minute papers are an effective way of involving all students in class simultaneously. It ensures equal participation of each and every class member, including anyone who may be too shy or fearful to participate orally. It can promote class attendance and attentiveness. It can improve the quality of class discussion by having students write briefly about a concept or issue before they begin discussing it.

- **Challenges:**

After teaching this particular topic students asked some time to prepare. So, I gave few minutes to prepare which takes more time to implement this activity than planned.

Changes required for future (if required):Nil

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	1	1	-	-	-	-	-	1	1	1	-	-

CO2: The students will be able to analyze the bio-chemicals and various physiological information

References:

4. <https://oncourseworkshop.com/self-awareness/one-minute-paper/>
5. <https://www.humber.ca/centreforteachingandlearning/instructional-strategies/teaching-methods/classroom-management-designing-the-environment/feedback-in-the-classroom/one-minute-papers.html>
6. M. Arumugam, "Biomedical Instrumentation", Second Edition.

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Department of Electronics and Communication Engineering

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Innovative Practices Description

UNIT III (ASSIST DEVICES)

Degree, Semester& Branch: V Semester B.E. ECE B

Course Code & Title: EC8073 & Medical Electronics

Name of the Faculty member: Mr.S.Vijayakumar

Name of the Topic: Magnetic Resonance Imaging Systems

Name of the Innovative Practice: Mind map

Date & Time: 27.08.2019 & 15 Minutes

ICT Tool Used: Black Board

Description:

A mind map is an easy way to brainstorm thoughts organically without worrying about order and structure. It allows the student to visually structure the ideas which is used to help with analysis and recall.

A mind map is a diagram for representing tasks, words, concepts, or items linked to and arranged around a subject using a non-linear graphical layout that allows the student to build an intuitive framework around a central concept.

Goals (Learning Outcomes):

- The students will be able to identify and visually record current understandings.
- The students will be able to summarize key information, clarify relationships or associations between information and ideas and draw conclusions.

Use of appropriate methods:

Justification for choosing the topic using mind map activity:

Magnetic Resonance Imaging (MRI) is has emerged as a powerful imaging technique in the medical field because of its high resolution capability and potential for chemical specific imaging. MRI consists of many building blocks such as Transmitter, receiver, Gradient coil and many technical terms such as Magnetic moment, free induction decay, Relaxation time. Mind map activity helps the students to easily grasp the overview of whole operations of MRI with the help of graphical representation of MRI building block.

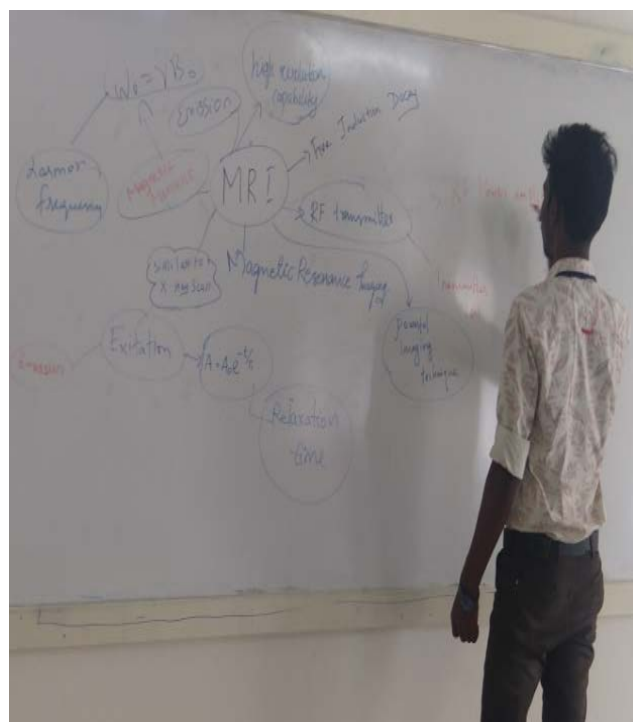
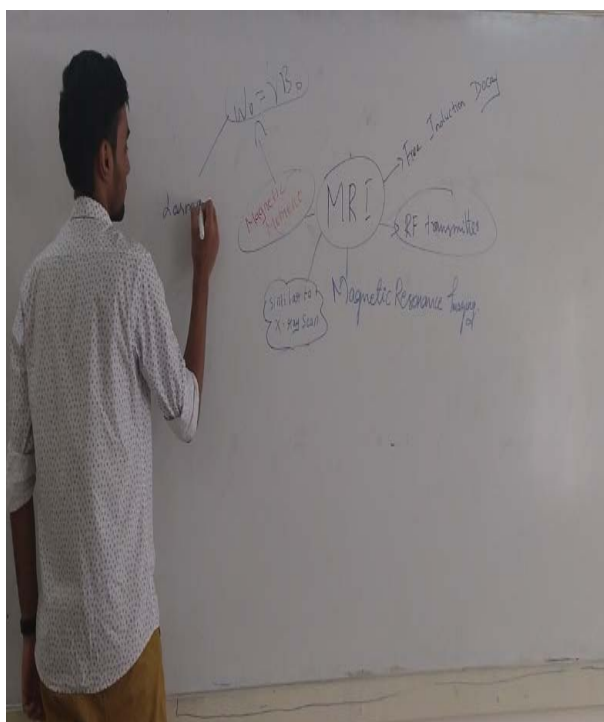
Effective presentation

Details of Implementation:

I explained to students about the strategy and ensure that students understand that mind maps are personal representations and as such they are not 'right' or 'wrong'. I Select a topic Magnetic Resonance Imaging Systems and write this in the centre of a blackboard. Students then identify connected key words or phrases related to Magnetic Resonance Imaging Systems and write these around the topic, progressively moving to less directly related words.

Innovative Teaching Method Execution

Magnetic Resonance Imaging Systems - Mind maps



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- Every students got equal opportunity to come forward to take part in this activity
- The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correctly.

Reflective Critique:

- **Benefits:**

Mind mapping increases creativity and productivity because it's an excellent tool to let students generate more ideas, identify relationships among the different data and information, and effectively improve students memory and retention. Making a mind map is an excellent way for students to be able to sort through your thoughts and ideas. This activity allows the students to quickly generate creative and even unique ideas in less time. With the help of this activity the students can be able to easily differentiate between different building blocks of MRI.

- **Challenges:**

I have planned this activity for 10 minutes but few students not able to recall every details on this topic. So, I gave them another 10 minutes to prepare.

Changes required for future (if required): Nil

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	1	1	-	-	-	-	-	1	1	1	-	-

CO3: The students will be able to describe various assist devices used in hospitals

References:

1. <https://www.ayoa.com/imindmap/articles/mind-mapping-for-innovation/>
2. Khandpur, R.S., "Handbook of Biomedical Instrumentation", TATA Mc Graw-Hill, New Delhi, 2003

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UNIT-IV(PHYSICAL MEDICINE AND BIOTELEMETRY)

Degree, Semester& Branch: V Semester B.E. ECE B

Course Code & Title: EC8073 Medical Electronics

Name of the Faculty member: Mr.S.Vijayakumar

Name of the Topic: Diathermy

Name of the Innovative Practice: Animation Video

Date & Duration: 13.09.2019& 10 minutes

ICT Tool Used: Desktop, projector

Description:

Animation is a brilliant and innovative new way to encourage students to communicate stories, ideas and concepts in a creative and original way. It can be particularly useful as a tool to encourage the creativity of students. Animated videos are a one stop solution that can grab the attention of the students. With its entertaining visuals, it can clearly comprehend the concept. Since it is an analogy to cartoons, students will get hooked to it. By using animation videos, students learn analytical thinking as to how to present a scenario.

Goals (Learning Outcomes):

- The students will be able to easily identify the different types of electrodes used for diathermy
- The students will be able to explain the operation of diathermy

Use of appropriate methods:

Justification for choosing the topic using animation video activity:

Diathermy means “through heating” or producing deep heating directly in the tissues of the body. The heat is applied with the help of electrodes. Animated video helps to show how the heat is developed in the electrode and the same is applied to the body to heal the wounds.

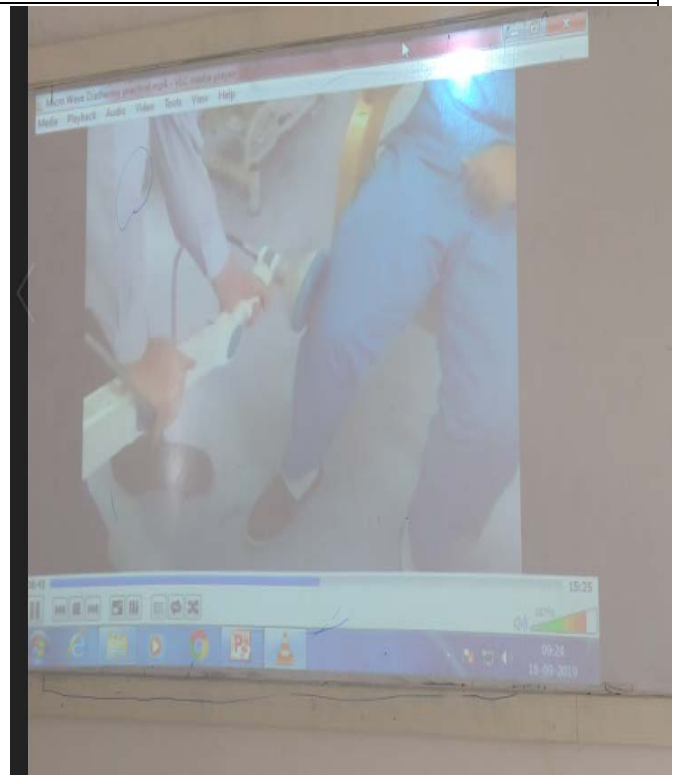
Effective presentation

Details of Implementation:

The suitable animated video for diathermy was displayed to the students and explained about the different types of diathermy and their procedure on how to apply to the human body.

Innovative Teaching Method Execution-UNIT-IV

Diathermy - Animated video



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- The success of the activity was evaluated by asking the same question in Internal Assessment test III – Around 80% of students answered correctly.

Reflective Critique:

- **Benefits:**

The students show more interest by listening to animated video, apart from theory concept. Animation videos for diathermy help the students to know how the diathermy helps to heal the tissues and wounds. It helps to differentiate the different types of diathermy and their related procedure on applying to patients.

- **Challenges:**

It takes some time to choose relevant videos for the diathermy topic in order to make understandable to students.

Changes required for future (if required):Nil

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO4	1	1	-	-	-	-	-	1	1	1	-	-

CO4: The students will be able to explain the equipment used for physical medicine and the various recently developed diagnostic and therapeutic techniques

Reference:

1. <https://stonesoup.com/post/why-animation-is-important/>
2. Khandpur, R.S., "Handbook of Biomedical Instrumentation", TATA McGraw-Hill, New Delhi, 2003.

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Innovative Practices Description

UNIT V (RECENT TRENDS IN MEDICAL INSTRUMENTATION)

Degree, Semester & Branch: V Semester B.E. ECE B

Course Code & Title: EC8073 & Medical Electronics

Name of the Faculty member: Mr.S.Vijayakumar

Name of the Topic: Lab on a chip

Name of the Innovative Practice: Brainstorming

Date & Time: 03.10.19 & 30 Minutes

ICT Tool Used: -

Description:

Brainstorming is a group creativity technique by which efforts are made to find a conclusion for a specific problem by gathering a list of ideas spontaneously contributed by its members. All the ideas are noted down without criticism and after the brainstorming session the ideas are evaluated.

Goals (Learning Outcomes):

- The students will be able to develop creative thinking processes and problem solving skills.
- The students will be able to recall existing knowledge and organize ideas

Use of appropriate methods:

Justification for choosing the topic using Brainstorming activity:

Lab on a chip is miniaturized integrated circuits which consists of sensors, transducers, amplifiers which can measure or test pH level, blood pressure, blood flow rate simultaneously and give the output etc., The students already learned this concept in previous chapter. So I asked them to prepare and share their ideas.

Effective presentation

Details of Implementation:

The process begins with a well-defined topic. Students are divided into groups. Each participant in a group creates an individual brainstorm around the topic. Each group need to present their views on stage. All the ideas are then merged into one large idea map. During this consolidation phase the participants may discover a common understanding of the issues as they share the meanings behind

their ideas. As the sharing takes place, new ideas may arise by the association. Those ideas are added to the map as well. Then ideas are generated on both individual and group levels.

Innovative Teaching Method Execution

Lab on a chip - Brainstorming



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- The success of the activity was evaluated by asking the same question in Internal Assessment test III – Around 80% of students answered correctly.
- Slow learners came forward to explain their ideas.

Reflective Critique:

• Benefits:

Brainstorming builds involvement, commitment, loyalty, and enthusiasm. Participating in the sessions stimulates and unlocks people's creative talents. Brainstorming also builds self-esteem because people are being asked for their participation and their ideas.

• Challenges:

Each group allotted for 5 minutes to present their ideas on stage. But few groups take more than the time allotted.

Changes required for future (if required):Nil

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	1	1	-	-	-	-	-	1	1	1	-	-

CO5: The students will be able to apply the concepts of medical Instrumentation in recent technology (Radio pill, Telemedicine, Endomicroscopy unit)

References:

1. <https://business.tutsplus.com/articles/top-brainstorming-techniques--cms-27181>
2. Khandpur, R.S., "Handbook of Biomedical Instrumentation", TATA Mc Graw-Hill, New Delhi, 2003.